

PRIYANSHU ROY

AI Systems Engineer & Machine Learning Researcher

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SUMMARY

Third-year Artificial Intelligence & Machine Learning undergraduate specializing in Large Language Models (LLMs) and Reinforcement Learning (RL). Dedicated to engineering production-ready AI products with a strong focus on automation, user engagement, and secure system design. Proven track record of guiding cross-functional projects from concept to deployment, delivering scalable architectures that drive measurable business impact while maintaining strict data privacy.

TECHNICAL SKILLS

Languages: Python, Rust, TypeScript, Java, Bash Scripting.

Machine Learning & AI: PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, OpenCV, Neural Networks, Reinforcement Learning, Generative Adversarial Networks, Explainable AI.

LLMs & Agentic AI: Supervised Fine-Tuning, Preference Optimization, Unsloth, vLLM, LangGraph, Retrieval-Augmented Generation, Vector Databases (LanceDB, pgvector), Prompt Engineering.

Edge AI & Systems: Open Neural Network Exchange (ONNX), Quantization-Aware Training, Post-Training Quantization, Android Native Development Kit, Java Native Interface, On-Device Inference.

Simulation & Robotics: NVIDIA Isaac Lab, Omniverse PhysX, Inverse Kinematics, Continuous Control, Multi-Agent Coordination.

Infrastructure & Tools: Next, React, Tauri, FastAPI, PostgreSQL, Supabase, Docker, Git, Continuous Integration and Continuous Deployment, Linux Architecture.

TECHNICAL EXPERIENCE & PROJECTS

RevExBot | Deep Reinforcement Learning Framework

March 2026 – Present

Continuous Control Robotics | Tech Stack: PyTorch (PPO/GAIL), NVIDIA Isaac Lab, Omniverse PhysX

- **Engineered** a physically simulated RL framework contrasting Adversarial Skill Embeddings and Mixture of Experts, orchestrating **8,192** parallel environments at **50Hz**. Prioritized landscape generalization over rapid convergence by replacing AdamW with SGD + Nesterov Momentum, integrating PyTorch Automatic Mixed Precision to execute full Phase 2 GAN training with **40%** reduced epoch times.
- **Architected** a Universal Motion Model utilizing a causal CNN history encoder to prevent look-ahead leakage across **15+** robotic tasks, alongside a custom Generalized Advantage Estimation (GAE) "Teleportation Mask" that explicitly hid environment resets from the discriminator to maintain Lipschitz continuity and eliminate gradient collapse.

SSENSE | DPDP Alignment Pipeline & SLM Engine

June 2026 – Present

Machine Learning & Governance | Tech Stack: Unsloth, TRL, vLLM, Qwen3.5, TypeScript, VS Code API

- **Spearheaded** an end-to-end ML pipeline to audit Digital Personal Data Protection Act compliance by deploying an adversarial Qwen2-72B model to generate **8,000** synthetic training pairs (**95%** validity). Optimized a Qwen3.5-9B model utilizing Unsloth Triton kernels to scale context length to **24,576** tokens, avoiding OOM panics on a 128GB Grace Blackwell UMA by halving micro-batch sizes.
- **Systematized** automated VRAM Airlocks via Bash scripts to dynamically clear **75GB** of FP8 weights, cutting alignment time by **60%**. Engineered a custom VS Code "Serverless Wrapper" extension to seamlessly bridge local IDEs with hybrid GPU/CPU runtimes, entirely bypassing local hardware bottlenecks during iterative model inference to improve latency.

- **Built** a fully offline, multi-modal Retrieval-Augmented Generation (RAG) agent utilizing a Tauri frontend and a hybrid vector-graph knowledge store, eliminating cloud API latency entirely to enable secure semantic search across **10,000+** documents without data exfiltration.
- **Devised** a GPU-aware dynamic VRAM swapping strategy to orchestrate local Python microservices and seamlessly alternate between vision and language models, ensuring stable operation within strict **8GB** hardware constraints and achieving zero Out-Of-Memory crashes.

- **Constructed** a fault-tolerant LangGraph state machine to autonomously triage citizen grievances, processing **1,000+** multimodal submissions via RAG to reduce manual processing overhead by **85%**.
- **Architected** an automated Groq-to-Gemini failover system functioning as a circuit breaker, enforcing strict DPDP compliance with dynamic data redaction to achieve **99.8%** routing precision.

- **Deployed** an edge-native ML firewall on Android utilizing **3** synchronized INT8-quantized TFLite models, engineering memory-safe custom arena allocators that prevented all OOM crashes during sustained traffic bursts.
- **Forged** a native C++/JNI bridge to the OS-level VpnService for zero-copy, zero-exfiltration deep packet inspection, translating raw flow heuristics into human-readable forensics via SHAP-based Explainable AI with **<50ms** inference latency.

CERTIFICATIONS

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| • IBM AI Engineering Professional Certificate – IBM | 2025 |
| • Goethe-Zertifikat A1: Start Deutsch 1 – Goethe-Institut | 2023 |

ACHIEVEMENTS

- **National Space Hackathon 2026 (IIT Delhi):** Designed the **Autonomous Constellation Manager (ACM)**, forulating autonomous coordination loops and multi-agent tracking for satellite constellations.
- **Algorithmic Rigor:** Solved **300+** complex problems on LeetCode and HackerRank, covering data structures, dynamic programming, and systems-level memory optimization.
- **Open Source Contributor:** Contributed to public repositories by shipping features, optimizing execution latency, and resolving runtime bugs in edge-AI and security tools across **5+** projects.

EDUCATION

Institute of Engineering and Management (IEM)	Kolkata, India
Bachelor of Technology in Artificial Intelligence & Machine Learning CGPA: 8.7/10.0	2024 – 2028
Calcutta Public School	Kolkata, India
Class XII (ISC) – Computer Science 88%	2024
Class X (ICSE) – Science 93%	2022